Total variation → Does not capture proximity.

Eg → $X+a$ → can go close 1. But Total variation does not capture it.

It's equal to 1, as long as $a \neq 0$.

→ It does not capture horizontal movements.

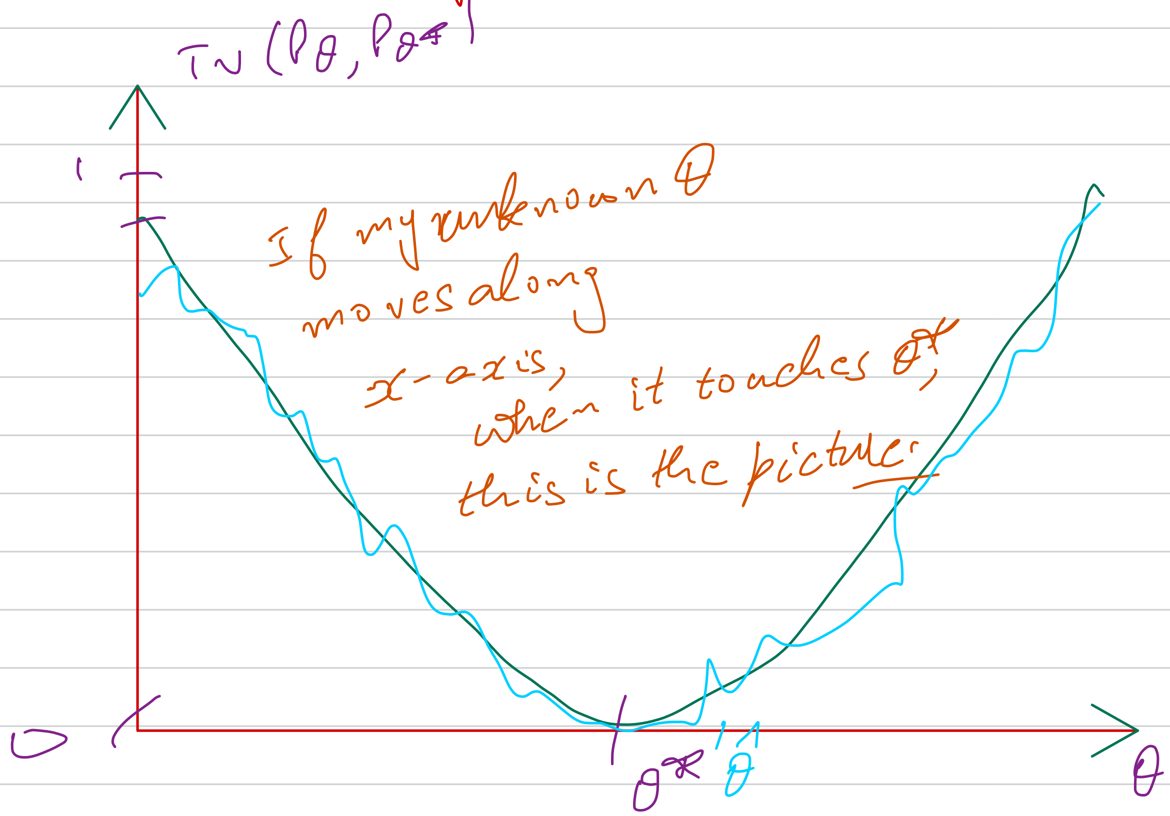
Eg → when I take my R.V., and move it a little bit, TV stops paying attention when they are disjoint (I think area under the curve)

$$\frac{1}{2}\left[1 + \quad 1 - e^{-\rho}\right]$$

$$\frac{1}{2}\left[\rho - e^{\rho}\right]$$

$e^{-\rho \cdot b}$

KL- Divergence:



$\theta = \theta^*$, area under the curve = 0.

→ If my estimator is good, my TV would be close θ^*, MINIMUM OF TV = θ^*.

we can suggest $\hat{\theta}$, to be the stimator of θ^*.

→ This θ is the principle behind every estimator we are going to talk about.

→ If you have a fn. that is minimized at the parameter you are looking for, try to estimate that fn.

AND MINIMIZE THE ESTIMATOR INSTEAD.

Unclear how to build an estimator for total variation → It becomes trivial

How? Let's see methods for estimation.

Then, I am saying, I need to estimate something.

How do I do this?

MAXIMUM LIKELIHOOD ESTIMATION OF THE TOTAL VARIATION.

↓ what do you do?

Consistent Replacing the avg.

↓

Like we did for the KISS example.

↓

That's our building block.

(The LLN is a good guiding principle) → This fn if I can express as something that's minimized at θ^*

As the expectation of something

$\phi\left(\frac{X-\mu}{\sigma} < \mu\right) = 0.5$

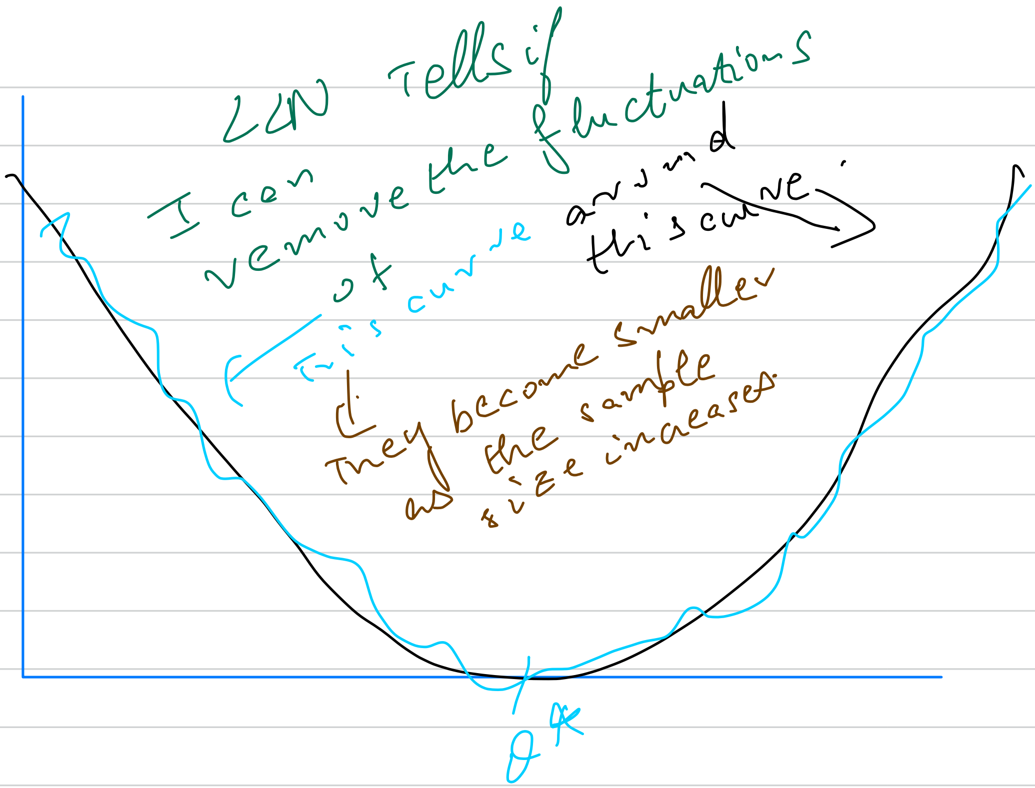
$P(X < \mu) = 0.5.$

$\phi(\mu) = 0.5.$

And you can replace the expectation by avg.

$|0.5 - 1|$

Now you have an estimator.



Replaces the TV, by another notion of distance between probability measures, but

Expectation of something.

I can say

(Let me replace the Expectation by avg.)

Kullback-Leibler Divergen

$\rightarrow$ Relative Entropy

Twice

Asymmetric

once

Information

Theory

$$TV = \frac{1}{2} \int \left| f_{\theta}(x) - f_{\theta^*}(x) \right| dx$$

$\downarrow$

L1-norm

Def^n of K-L divergence:

Let P and Q be discrete probability distributions with pmfs p and q respectively.

$P, Q \in E$.

KL-divergence (also known as relative entropy) between P and Q is defined by.

$$KL(P,Q) = \sum_{x \in E} p(x) \left(\ln \left(p(x) \right) - \ln q(x) \right)$$

$a \ln(b) + (1-a) \ln(1-b)$.

$$= \sum_{x \in E} p(x) \ln p(x) - \sum_{x \in E} p(x) \ln(q(x))$$

Sum is only the support of P.

$$= \int_{x \in E} p(x) \ln \left(\frac{p(x)}{q(x)} \right).$$

where this integral is only over the support of P.

Why only over P?

Any point outside support of P where $q(x) \neq 0$. ($x \in E$)

$$\lim_{p/q \to 0^+} q\left(\frac{p}{q}\right)\ln\left(\frac{p}{q}\right) = q \lim_{p/q \to 0^+} \left(\frac{p}{q}\right)\ln\left(\frac{p}{q}\right)$$

$$= q \cdot (0) = 0.$$

(By L'Hospital's Rule).

It is not a distance, it's a divergence.

Properties of $\boxed{\text{KL-divergence}}$:

→ (Lots) (lot of hacks)

▷ $KL(P_\theta, P_{\theta'}) \neq KL(P_{\theta'}, P_\theta)$ in General.

▷ $KL(P_\theta, P_{\theta'}) \geq 0$.

▷ If $KL(P_\theta, P_{\theta'}) = 0$, then ($P_\theta = P_{\theta'}$ definite).

▷ $KL(P_\theta, P_{\theta'}) \not\leq KL(P_\theta, P_{\theta''}) + KL(P_{\theta''}, P_\theta)$ in General.

→ Attempts to make it symmetric by summation over $p(x)$ and $q(x)$

Not getting much apart from symmetry.

Shanon-Jensen's divergence

Deep Neural Nets.

Symmetric → Nothing else.

→ Jensen's Inequality → Log is concave,

If applied to KL-divergence

It will show it is +ve.

→ It is definite.

Definite:

INTUITION:

(The fact that it's only tells me it's 0 at θ^*).

If those curves touch each other, any minimizer, I come up with could be one of those guys.

For TV, the only thing we used was the black curve would touch the x-axis was at θ^*

Model Identifiable

θ^* is the unique minimizer of
$$\theta \to KL(P_{\theta^*}, P_\theta).$$

(Even if it's not symmetric, it's still the unique minimizer)

$\to$ Identifiable models.

ESTIMATING THE K-L DIVERGENCE:

1 MAXIMUM LIKELIHOOD ESTIMATION:

→ So, same estimation strategy for the new estimator.

K-L.

Good notion :: if I can find something that makes θ^* to 0

Identifiability.

$$a\left(\ln\left(\frac{b}{1-b}\right) - a\right.$$

$$a(\ln a - a)\ln(b).$$

<

Estimating KL-divergence:

Need to find an Estimator → MLE

→

INTUITION:

→ KL is a good notion

→

∴ It finds something which makes it equal to 0.

I know I found the right distribution

(Identifiability)

(Right parameter)

STARTED WITH THIS IDEA

KL divergence

$$KL(P_{\theta^*}, P_\theta) = E_{\theta^*}[\log P_{\theta^*}(x)] - E_{\theta^*}[\log p_\theta(x)]$$

why useful!

How ?

$$KL(P_{\theta^*}, P_\theta) = \sum p_{\theta^*} \log\left(\frac{p_{\theta^*}(x)}{p_\theta(x)}\right)$$

$$= \sum p_{\theta^*} \log(p_{\theta^*}(x)) - \sum p_{\theta^*} \log(p_\theta(x)).$$

$$= \underbrace{\sum p_{\theta^*} H_{p_{\theta^*}}(x)}_{} - \sum p_{\theta^*} H_{p_\theta}(x);$$

$\theta^* \to p_{\theta^*}$ (definition of Expectation) $\theta^* \to p_\theta$.

$$= E_{p_{\theta^*}}[H_1] - E_{p_{\theta^*}}[H_2].$$

$$= \underbrace{E_{p_{\theta^*}}[\log p_{\theta^*}(x)]}_{\text{constant.}} - E_{p_{\theta^*}}[\log p_\theta(x)].$$

Def^n:

$$\boxed{E[g(x)] = \sum p(x)\, g(x).}$$

$$= \underbrace{E_{p_{\theta^*}}[\log p_{\theta^*}]}_{\text{(constant)}} - E_{p_\theta}[\log p_\theta].$$

Remember: My one and only estimation strategy right now, is to replace expectation by averages.

$$= \left(\underbrace{\text{CONSTANT}_{p_{\theta^*}}}_{\text{This would change}}\right) - E_{\theta^*}[\log p_\theta(x)].$$

(The value where it is minimized won't change)

I have data from this Expectation.

(I have data from p_{θ^*}.)

Only Trick! LLN.

Replace it with avg.

$$E_{\theta^*}[h(X)] \rightarrow \frac{1}{n}\sum_{i=1}^{n}(h(X_i)) \text{ by LLN.}$$

$$\therefore \widehat{KL}(P_{\theta^*}, P_\theta) = \text{"constant"} - \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(x_i).$$

Deriving the MLE:

→ (1) $\min_{\theta \in \Theta} \widehat{KL}(P_{\theta^*}, P_\theta) \iff \min_{\theta \in \Theta} -\frac{1}{n}\sum_{i=1}^{n} \log p_\theta(x_i)$

(This means the θ that minimizes LHS, is the same as RHS).

(Just ∵ I am shifting up and down the curve so whatever the argument was, it just won't move)

→ Just minimizing the negative

which I guess would be maximizing

(n^{th} order statistics)

$$\iff \max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \log(x_i).$$

$$\iff \max_{\theta \in \Theta} \frac{1}{n} \log\left(\prod_{i=1}^{n} p_\theta(x_i)\right)$$

log is an increasing fn.

$$\iff \max_{\theta \in \Theta} \prod_{i=1}^{n} p_\theta(x_i).$$

minimizer.

This is the maximum LIKELIHOOD PRINCIPLE.

Remarks:

1. The quantity $\hat{\theta}_n =$ maximizer of $\prod_{i=1}^{n} p_\theta(x_i)$. is referred to as the minimum likelihood estimator.

NOTE:

→ This is the same as the estimator $\hat{\theta}_n =$ minimizer of $-\frac{1}{n} \sum_{i=1}^{n} [\log p_\theta(x_i)]$.

Remark 2: Under certain technical conditions, the maximum likelihood estimator is guaranteed to converge to the true parameter θ^*

Likelihood Estimator:

→ A likelihood of a model,

→ Discrete E →

↳ It's actually a n-sample and a candidate parameter.

Definition:

Likelihood can be described for a given

Likelihood of a model is defined as:

$L_n : (E^n \times \Theta \to \mathbb{R})$ → map: L_n or just L

$(x_1, \dots x_n, \theta) \to P_\theta[X_1 = x_1, \dots, X_n = x_n]$

MEANING OF LIKELIHOOD → (How likely this data was generated by P_θ)

→ Probability of likelihood.

~~Let's~~ ~~look at the~~ ~~probability~~ that those numbers were coming from θ.

$xp + (1-p) p$.

(the bigger n) $(1-p)$

The bigger this number (x) $(1-x)$ the more likely it is.

p $1-p$

$\{X = x_1, X = x_2 \dots X = x_n\}$ $x=1$ $x=0$

$p + 1-p$, $p + p + p$.

That's what likelihood means.

$(1-p) + (1-p)$

$(1-p)(1-p)$.

(for different values of θ)

$E^m \times \theta \longrightarrow R$

→ 2 candidate values ; $\theta = 0$ or $\theta = 1$.

→ Which one is the more likely to have generated this data.

↳ More θ.

Dual Vision of Statistics And Probability.

(I am just computing the probability of having seen these numbers for different values of θ)

(Fix x_1 → Compute Probability) θ is a fixed

SIMPLER WAY TO THINK

To values of $\theta \rightarrow$

$\theta \rightarrow 0$
$\theta \rightarrow 1$

$\rightarrow$ You could try which one is the more likely to having generated your data

---

Example: $Ber(p)$.

$\hookrightarrow$ Either $Ber(p) = \{0\}$

$Ber(p) = \{1\}$.

when $p = 0$

True parameter

Now, n samples:
$0, 0, 0, 0, \ldots$

Next page

Question: what is the prob; $x_1 = 0, x_2 = 0, \ldots, x_n = 0$

$\therefore Ber(0) = 1$.

if $1, 1, 1, 1, \ldots$ $p = 0$

$Ber(0) = 0$.